C Programming Assignment – Question 5

5. Car Mileage

Question

Given the car's mileage, available petrol, and distance remaining, determine whether the car can reach its destination. Print “Can reach” or “Cannot reach”.

Sample Input

```
11
6
67
```

Sample Output

```
Cannot reach
```

C Program

```
#include <stdio.h>

int main(void) {
    float mileage;
    int petrol, distance;

    printf("Enter mileage (km per litre): ");
    scanf("%f", &mileage);

    printf("Enter petrol available (litres): ");
    scanf("%d", &petrol);

    printf("Enter distance to destination (km): ");
    scanf("%d", &distance);

    if (mileage * petrol >= distance) {
        printf("Can reach\n");
    } else {
        printf("Cannot reach\n");
    }

    return 0;
}
```

Explanation

1. Calculate the possible distance using mileage × petrol.

2. Compare the possible distance with the required distance.

3. Print “Can reach” if the possible distance is enough; otherwise print “Cannot reach”.

Note

11 km/litre × 6 litres = 66 km, which is less than 67 km.